



QR Code for
report verification

Case ID	: 2600130362	Sample Type	: FFPE TISSUE BLOCK
Name	: MR. AGATAMUDI VENKATA KAMANA (RJM2604120002)	Date & Time Collected	: 12-Apr-2026 12:00 AM
Sex/Age	: Male/61 Years	Date & Time Received	: 15-Apr-2026 04:44 PM
Bill. Loc.	: OncQuest Laboratories Ltd., New Delhi	Date & Time Reported	: 28-Apr-2026 06:16 PM
Ref. By	:	Report Version	: 1
Indication	:		

Brainseq Genomic Profiling Panel – Advance
On Illumina Novaseq 6000 Platform

Clinical Indication:

Features are consistent with recurrent astrocytoma, IDH - mutant, CNS WHO grade 4
Block No.: 24-4733-B2

Positive

(Clinically significant Variant detected)

Report Highlights:

Tier I	Tier II	Tier III	MSI	Therapies	Clinical Trials
2	0	0	Stable	2	18

Key Findings:

Tier	Gene & Transcript	Variant	Exon	Coverage/VAF
IA	IDH1 NM_005896.4	c.395G>A p.(R132H)	4	890x / 47.19%
IA	TERT NM_198253.3	c.-124C>T (Upstream Mutation)	-	410x / 36.36%

*Genetic test results are reported based on the recommendations of AMP-ASCO-CAP guidelines.

**Public data sources included in relevant targeted therapies, guidelines: EMA, FDA, ESMO, NCCN.

Reviewed by **PRAKASH PATEL**

Dr. Salil Vaniawala
M.Sc Ph.D.(Human Genetics)
Consulting Geneticist

Dr.Nirmal A. Vaniawala
MD (Path. & Bact.)

Dr. Salil Vaniawala
Ph.D.(Human Genetics)
Consulting Geneticist



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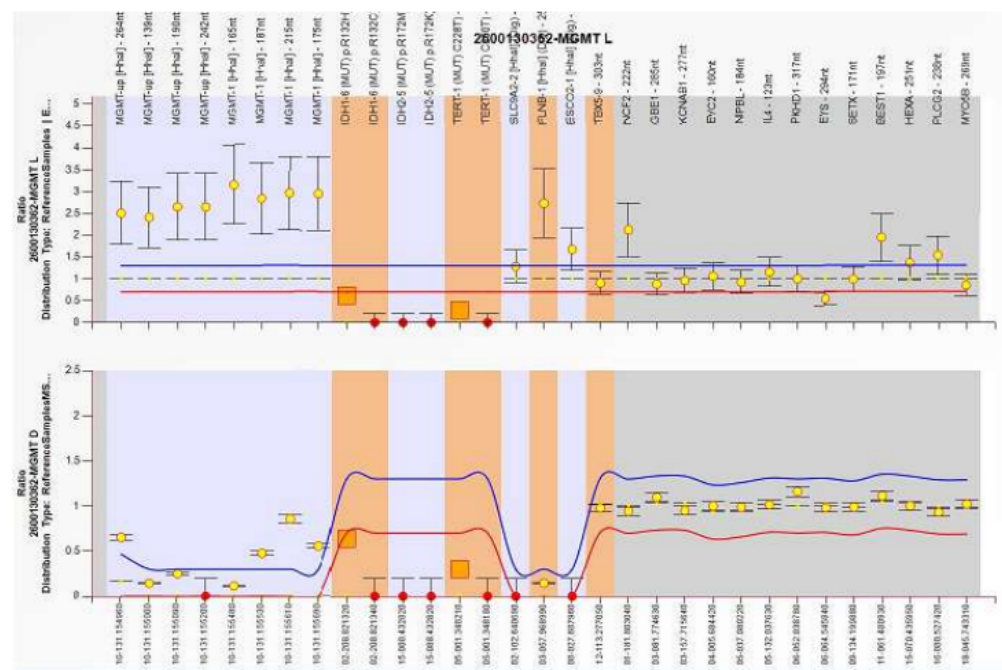
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Biomarkers Results:

Assay Biomarker	Result	Interpretation
Microsatellite Instability (MSI)	Stable	Stable Microsatellite detected
MGMT Methylation Status	Positive	MGMT Methylation and TERT C228T, IDH1 R132H Mutation Detected
FISH 1p19q Co-Deletion	Positive	nuc ish(1p36×1,1q25×2)[45] / nuc ish(1p36×2,1q25×2)[05] nuc ish(19q13×1,19p13×2)[48] / nuc ish(19q13×2,19p13×2)[03]

MLPA Findings:

Data from Clinical Sample



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[Signature]

Dr. Salil Vaniawala
M.Sc Ph.D.(Human Genetics)
Consulting Geneticist

[Signature]

Dr.Nirmal A. Vaniawala
MD (Path. & Bact.)

[Signature]

Dr. Salil Vaniawala
Ph.D.(Human Genetics)
Consulting Geneticist

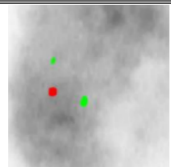
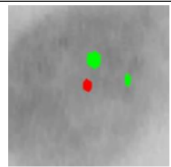


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FISH Findings:

Data from Clinical Sample

NO. OF CELLS STUDIED: 50				BLOCK No.: SP24 -4733 B2		
	CHROMOSOME	1p36 ORANGE	1q25 GREEN	RATIO	RESULT	INTERPRETATION
	NO. OF SIGNALS	1	2	0.55	DELETED	POSITIVE for co-deletion if ratio <0.80
RESULT: nuc ish(1p36×1,1q25×2)[45] / nuc ish(1p36×2,1q25×2)[05]						
	CHROMOSOME	19q13 ORANGE	19p13 GREEN	RATIO	RESULT	INTERPRETATION
	NO. OF SIGNALS	1	2	0.52	DELETED	POSITIVE for co-deletion if ratio <0.80
RESULT: nuc ish(19q13×1,19p13×2)[48] / nuc ish(19q13×2,19p13×2)[03]						
INTERPRETATION :				POSITIVE		

Clinical Significance:

Variant	Therapy	
	In This Cancer	In Other Cancer
IA IDH1 p.(R132H)	- ivosidenib - vorasidenib	- ivosidenib - vorasidenib

Variant	Diagnostic Significance	Prognostic Significance
IDH1 p.(R132H)	Astrocytoma IDH-mutant (Grade 4)	NCCN: Favorable

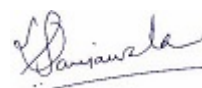
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M.Sc Ph.D.(Human Genetics)
Consulting Geneticist



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Biomarker Description:

Variant	Biomarker Description
IDH1 NM_005896.4 chr2:209113112 c.395G>A p.(R132H)	The IDH1 and IDH2 genes encode homologous isocitrate dehydrogenase enzymes that catalyze the conversion of isocitrate to α-ketoglutarate (α-KG)[1]. The IDH1 gene encodes the NADP+ dependent cytoplasmic isocitrate dehydrogenase enzyme; IDH2 encodes the mitochondrial isoform[1]. Recurrent somatic mutations in IDH1 and IDH2 are mutually exclusive and observed in several malignancies, including glioma, chondrosarcoma, intrahepatic cholangiocarcinoma, acute myeloid leukemia (AML), and myelodysplastic syndrome (MDS)[2]. Recurrent IDH1 variants include predominantly R132H/C plus other substitutions at lower frequencies[3]. These gain-of-function variants confer neomorphic enzyme activity[4]. Although wild-type enzymatic activity is ablated, recurrent IDH1 variants catalyze the conversion of α-KG to D-2-hydroxyglutarate, an oncometabolite with diverse effects on cellular metabolism, epigenetic regulation, redox states, and DNA repair[1,5]. Recurrent IDH1 mutations are present in 5-10% of patients with AML and 5% of patients with MDS[6,7,8]. Recurrent IDH1 mutations are present in nearly 80% of lower-grade diffuse gliomas[9,10]. Alterations in IDH1 are rare in pediatric cancers[9,10]. Somatic mutations in IDH1 are observed in 2% of glioma and less than 1% of leukemia (2 in 311 cases), B-lymphoblastic leukemia/lymphoma (1 in 252 cases), embryonal tumor (1 in 332 cases), and Wilms tumor (1 in 710 cases)[9,10]. The IDH1 and IDH2 inhibitor vorasidenib[11] is FDA-approved (2024) for the treatment of adults and children with Grade 2 astrocytoma and oligodendroglioma with IDH1 R132C/G/H/L/S mutations. Additionally, olutasidenib[12] (2022) and ivosidenib[13] (2018) are FDA-approved for treating IDH1 R132C/G/H/L/S variants in AML. Ivosidenib is also approved for treating myelodysplastic syndrome (MDS) and cholangiocarcinoma patients with the same IDH1 variants[14]. IDH1 mutations are associated with favorable outcome in lower-grade gliomas, astrocytoma, and oligodendroglioma with 1p/19 codeletion[15,16,17]. Conversely, IDH1 mutations are associated with inferior leukemia-free survival in primary myelofibrosis (PMF)[18,19]. Mutations in IDH1 are diagnostic of IDH-mutated astrocytoma and oligodendroglioma with 1p/19q-codeletion subtypes of central nervous system (CNS) tumors[15,20]. ivosidenib TIBSOVO® is an isocitrate dehydrogenase-1 (IDH1) inhibitor indicated for the treatment of adult patients with a susceptible IDH1 mutation as detected by an FDA-approved test with: Newly Diagnosed Acute Myeloid Leukemia (AML) • In combination with azacitidine or as monotherapy for the treatment of newly diagnosed AML in adults 75 years or older, or who have comorbidities that preclude use of intensive induction

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M.Sc Ph.D.(Human Genetics)
Consulting Geneticist

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MD (Path. & Bact.)

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chemotherapy.

Relapsed or refractory AML

• For the treatment of adult patients with relapsed or refractory AML.

Relapsed or refractory Myelodysplastic Syndromes (MDS)

• For the treatment of adult patients with relapsed or refractory myelodysplastic syndromes.

Locally Advanced or Metastatic Cholangiocarcinoma

• For the treatment of adult patients with locally advanced or metastatic cholangiocarcinoma who have been previously treated.

Cancer type: Cholangiocarcinoma

TERT
NM_198253.3
chr5:1295228
c.-124C>T
(Upstream
Mutation)

The TERT gene encodes telomerase reverse transcriptase, a component of the telomerase core enzyme along with the internal telomerase RNA template (TERC)[21]. TERT is repressed in most differentiated cells, resulting in telomerase silencing[21]. In cancer, telomerase reactivation is known to contribute to cellular immortalization[21,22]. Increased TERT expression results in telomerase activation, allowing for unlimited cancer cell proliferation through telomere stabilization[21]. In addition to its role in telomere maintenance, TERT has RNA-dependent RNA polymerase activity, which, when deregulated, can promote oncogenesis by facilitating mitotic progression and cancer cell stemness[21].

Alterations and prevalence Somatic mutations are observed in 4% of skin cutaneous melanoma and uterine corpus endometrial carcinoma, 3% of kidney renal papillary cell carcinoma, and 2% of pancreatic adenocarcinoma, stomach adenocarcinoma, and sarcoma[9,10]. Additionally, TERT promoter mutations causing upregulation are observed in many cancer types, especially non-aural cutaneous melanoma (80% of cases), and glioblastoma (70% of cases)[22]. Specifically, TERT promoter mutations at C228T and C250T are recurrent and result in de novo binding sites for ETS transcription factors, leading to enhanced TERT transcription[21]. Amplification of TERT is observed in 15% of lung squamous cell carcinoma, 14% of esophageal adenocarcinoma, 13% of adrenocortical carcinoma and lung adenocarcinoma, and 10% of bladder urothelial carcinoma, 9% of ovarian serous cystadenocarcinoma, 6% of cervical squamous cell carcinoma, 5% of liver hepatocellular carcinoma, sarcoma, skin cutaneous melanoma, stomach adenocarcinoma, head and neck squamous cell carcinoma, 4% of uterine carcinosarcoma, 3% of uterine corpus endometrial carcinoma, breast invasive carcinoma, and 2% of diffuse large B-cell lymphoma[9,10]. TERT is overexpressed in over 85% of tumors and is considered a universal tumor associated antigen[23]. Alterations in TERT are rare in pediatric cancers[9,10]. Somatic mutations are observed in less than 1% of B-lymphoblastic leukemia/lymphoma (2 in 252 cases, Page 6 of 12 2 in 297 cases), bone cancer (1 in 327 cases), and Wilms tumor (1 in 710 cases)[9,10]. TERT amplification is observed in 1-2% of peripheral nervous

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Dr. Salil Vaniawala
M.Sc Ph.D.(Human Genetics)
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system cancers (2 in 91 cases), leukemia (2 in 250 cases), and B-lymphoblastic leukemia/lymphoma (5 in 731 cases)[9,10].

Potential relevance Currently, no therapies are approved for TERT aberrations. TERT promoter mutations are diagnostic of oligodendroglioma IDH-mutant with 1p/19q co-deletion, while the absence of promoter mutations combined with an IDH mutation is characteristic of astrocytoma[15,24]. Due to its immunogenicity and near-universal expression on cancer cells, TERT has been a focus of immunotherapy research, including peptide, dendritic, and DNA vaccines as well as T-cell therapy[23].

Therapy & Clinical Trials:

IDH1 p.(R132H) c.395G>A

Therapy	FDA	NCCN	EMA	ESMO	Trials
ivosidenib	✓	✓✓	✓	✓	
vorasidenib	✓	✓✓	X	X	
olaparib, durvalumab	X	X	X	X	II
olutasidenib, chemotherapy	X	X	X	X	II
supplement, retifanlimab	X	X	X	X	II
talazoparib	X	X	X	X	II
avutometinib, defactinib	X	X	X	X	I/II
AZD-9574, chemotherapy	X	X	X	X	I/II
NEO-212	X	X	X	X	I/II
ribociclib, chemotherapy	X	X	X	X	I/II
vorasidenib, chemotherapy	X	X	X	X	I/II
zotiraciclib	X	X	X	X	I/II
ABBV-706	Page 7 of 15 X	X	X	X	I

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amivantamab	X	X	X	X	I
ASKC-202	X	X	X	X	I
chemotherapy, pamiparib	X	X	X	X	I
MT-001 (Meton Pharmaceuticals)	X	X	X	X	I
MVR-C5252	X	X	X	X	I
tasadenoturev	X	X	X	X	I
TGFβR2KO/IL13Rα2 CAR T-cells	X	X	X	X	I

☐ Note: ✓ : In this Cancer Type, ✓ : In Other Cancer Type, ✓✓ : both In this and Other Cancer Type, X : No Evidence, △□ : Caution for Use

Test Description

Brainseq Genomic Profiling Panel - Advance is advancing precision oncology research through the analysis of multiple relevant biomarkers in a single next-generation sequencing (NGS) test.

The Brainseq Genomic Profiling Panel – Advance assay analyses **89** unique genes (**62** DNA + **27** RNA) for single gene and multiple gene biomarker insights, detects single gene biomarkers from all variant types including **SNVs, indels, CNVs, known and novel fusions, and splice variants**, analyze complex multi-gene biomarkers for mutational signatures, microsatellite instability (**MSI**) for immunotherapy research, **MGMT methylation** status & FISH for **1p19q co-deletion**.

The performance of the panel was verified using both commercially sourced reference controls and FFPE samples.

Test Methodology

Extraction & Library Preparation: DNA & RNA extracted from FFPE tissue block sample is used for targeted gene capture using a custom designed hybrid capture panel. In process quality controls were determined for prepared library.

Sequencing: The libraries were sequenced at mean depth: >1000x on Illumina Novaseq- next generation sequencing platform.

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Data Analysis: Sequencing data are analyzed using a customized and validated bioinformatics pipeline capable of detecting various classes of genomic alterations, including single nucleotide variants (**SNVs**), **Indels**, copy number variations (**CNVs**) and fusions. As part of the streamlined bioinformatics workflow, Multi-gene Biomarkers such as Microsatellite Instability (**MSI**) Score is calculated. Reportable genomic alterations and fusions are prioritized, classified, and reported based on AMP-ASCO-CAP guidelines and NCCN guidelines.

Limit of Detection (LOD): The LOD for SNVs and InDels is 5% Variant allele Frequency (VAF) and for Fusions is ≥ 10 supporting reads.

MLPA:

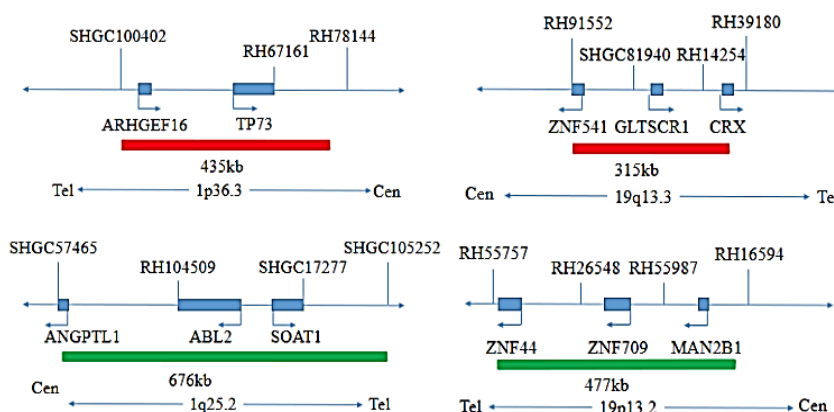
MGMT Methylation Analysis is based on the MS-MLPA technology (Methylation Specific Multiplex Ligation-dependent Probe Amplification) and employs the SALSA® MLPA® probe mixes available from MRC (Holland).

FISH:

1p36 and 19q13 – co-deletion By Fluorescence in situ Hybridization (FISH).

Healthcare 1p Dual color gene detection Probes Orange: 1p36, Green: 1q25.

Healthcare 19q Dual color gene detection Probes Orange: 19q13, Green: 19p13



Related Biomarker Information

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MGMT Methylation:

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Dr. Salil Vaniawala
M.Sc Ph.D.(Human Genetics)
Consulting Geneticist

Dr.Nirmal A. Vaniawala
MD (Path. & Bact.)

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Epigenetic silencing of the MGMT (O6-methylguanine–DNA methyltransferase) DNA repair gene by promoter methylation compromises DNA repair mechanism and has been associated with better response to treatment and longer overall survival in glioma patients treated with alkylating agents.

Biology of the process: MGMT is a DNA repair enzyme that reverses alkylation of guanine by transferring the alkyl group to the active site of the enzyme. Diminished MGMT expression due to methylation of CpG sites in the 5' portion of the MGMT gene allows accumulation of alkylguanine DNA which, following incorrect base pairing with thymine, triggers DNA damage signalling and cell death. MGMT methylation is associated with increased sensitivity of glioblastoma to alkylating agents such as BCNU (carmustine), and it also correlates with prolonged progression-free survival in glioma patients treated with temozolomide.

Clinical indications for MGMT promoter methylation testing include high grade glioma (anaplastic astrocytoma or glioblastoma) patient being considered for therapy with an alkylating agent, or glioma patient being considered for temozolomide therapy.

FISH 1p/19q co-deletion:

The presence of 1p/19q co-deletion predicts a better response to chemotherapy (such as temozolomide) and radiotherapy, which are standard treatments for oligodendrogliomas.

Patients with 1p/19q co-deletion generally have a better prognosis, meaning a longer survival and slower disease progression. The World Health Organization (WHO) classification of gliomas considers 1p/19q co-deletion status essential for a complete diagnosis of oligodendrogliomas.

Notes

Treatment decisions should not be based solely on the results of this test. Findings must be interpreted in the context of the patient's complete clinical profile—including pathological, radiological, and family history—to guide diagnosis, prognosis, and therapeutic decisions.

Therapy-related information in this report is derived from multiple sources including FDA-approved drug data, NCCN guidelines, peer-reviewed literature, standard clinical databases, and biomarker evidence strength. However, therapy options listed may not be suitable or beneficial for all patients. This report aims to provide a summary of potentially effective treatments, therapies that may pose increased risks, and those that may have limited benefit, based on genomic alterations identified. Prognostic and diagnostic biomarkers relevant to the disease context may also be reported.

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SN Genelab Pvt Ltd does not guarantee completeness, accuracy, or eligibility for trial enrollment. Physicians must verify eligibility criteria for specific trials independently.

Identification of a genomic alteration does not guarantee therapeutic efficacy or predict ineffectiveness. Final decisions on treatment should rest with the treating physician. The absence of reported therapeutic options does not imply ineffectiveness or lack of safety of any medication selected independently.

All variant classifications and related clinical information are based on the latest evidence from peer-reviewed publications, public clinical databases, and medical guidelines (e.g., NCCN, ASCO, AMP) available at the time of report generation. However, literature is continuously evolving, and classification may change with emerging evidence.

This test is performed on tumor tissue without a matched normal (e.g., blood) sample. Therefore, some variants may be of germline origin. While the assay is designed and validated for somatic variant detection, it does not differentiate between somatic and germline variants. If indicated, germline testing and genetic counselling are recommended.

FFPE-derived samples often pose challenges due to degradation. Poor-quality DNA or RNA may fail QC thresholds at various stages such as extraction, library preparation, or bioinformatics, and may result in assay failure or compromised results, such as low gene coverage or shallow variant depth. TMB score may be influenced by cytosine deamination artifacts (C>T transitions), especially in poorly fixed tissues.

Variants with allele frequencies between 3–5% may be reported if they are of strong or potential clinical relevance, or if previously detected in the same patient. False positives or negatives below assay sensitivity cannot be ruled out.

CNVs/gene amplifications have >80% detection sensitivity in this assay. Confirmation using orthogonal methods such as FISH/DDPCR is recommended when clinically indicated.

Variants in non-coding regions or deep intronic sequences are not assessed in this assay.

This assay is under the scope of NABL as Solidseq Comprehensive panel.

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DNA Gene List

AKT1	APC	ATRX	BAP1	BCOR
BCORL1	BRAF	CDK6	CDKN2A	CDKN2B
CIC	CTNNB1	DICER1	EED	EGFR
EPCAM	ERBB2	ERBB4	FGFR1	FGFR2
FGFR3	FUBP1	GNA11	GNAQ	H3F3A
HIST1H3C	HRAS	IDH1	IDH2	KDM6A
KRAS	MAP2K1	MET	MLH1	MSH2
MSH6	MYC	MYCN	NF1	NF2
NOTCH1	NRAS	PDGFRA	PIK3CA	PMS2
PTCH1	PTEN	RB1	SETD2	SF3B1
SMAD4	SMARCA4	SMARCB1	SMO	SRC
SUFU	SUZ12	TERT	TP53	TSC1
TSC2	VHL			

RNA Gene List

ALK	BRAF	CIC	EGFR	EML4
ETV6	EWSR1	FGFR1	FGFR2	FGFR3
FUS	KIAA1549	MAML2	MET	MN1
MYB	MYBL1	NTRK1	NTRK2	NTRK3
PRKCA	RAF1	ROS1	STAT6	TACC3
TFG	YAP1			

MSI

% of Unstable Loci

MSI - Stable	0-19%
MSI - Low / Intermediate	20-30%
MSI - High	>30%

MGMT Methylation

Methylation ratio of a probe in test sample > baseline	Methylation is increased
Methylation ratio of a probe in test sample ≤ baseline	Methylation is not increased

FISH 1p19q Co-Deletion

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Negative	If ratio is >0.80
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MD (Path. & Bact.)

Dr. Salil Vaniawala
Ph.D.(Human Genetics)
Consulting Geneticist



QR Code for
report verification

Case ID : 2600130362	Sample Type : FFPE TISSUE BLOCK
Name : MR. AGATAMUDI VENKATA KAMANA (RJM2604120002)	Date & Time Collected : 12-Apr-2026 12:00 AM
Sex/Age : Male/61 Years	Date & Time Received : 15-Apr-2026 04:44 PM
Bill. Loc. : OncQuest Laboratories Ltd., New Delhi	Date & Time Reported : 28-Apr-2026 06:16 PM
Ref. By :	Report Version : 1
Indication :	

Positive	If ratio is <0.80
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Tier	Interpretation
Tier IA	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to US FDA-approved therapies for a specific type of tumor or have been included in professional guidelines as therapeutic, diagnostic, and/or prognostic biomarkers for specific types of tumors.
Tier IB	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to a therapy based on well-powered studies with consensus from experts in the field, or have diagnostic and/or prognostic significance of certain diseases based on well-powered studies with expert consensus.
Tier IIC	Variants of Potential Clinical Significance: biomarkers that predict response or resistance to therapies approved by FDA or professional societies for a different tumor type (i.e., off-label use of a drug), serve as inclusion criteria for clinical trials, or have diagnostic and/or prognostic significance based on the results of multiple small studies.
Tier IID	Variants of Potential Clinical Significance: Biomarkers that show plausible therapeutic significance based on preclinical studies or may assist disease diagnosis and/or prognosis themselves or along with other biomarkers based on small studies or multiple case reports with no consensus.
Tier III	Variants of Unknown clinical significance (VUS): Not observed at a significant allele frequency in the general or specific subpopulation or pancancer or tumor specific variant databases. No convincing published evidence of cancer Association

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Reviewed by PRAKASH PATEL

Dr. Salil Vaniawala
M.Sc Ph.D.(Human Genetics)
Consulting Geneticist

Dr. Nirmal A. Vaniawala
MD (Path. & Bact.)

Dr. Salil Vaniawala
Ph.D.(Human Genetics)
Consulting Geneticist



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Dr. Salil Vaniawala
M.Sc Ph.D.(Human Genetics)
Consulting Geneticist

Dr.Nirmal A. Vaniawala
MD (Path. & Bact.)

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----- End Of Report -----

Reviewed by **PRAKASH PATEL**

Dr. Salil Vaniawala
M.Sc Ph.D.(Human Genetics)
Consulting Geneticist

Dr. Nirmal A. Vaniawala
MD (Path. & Bact.)

Dr. Salil Vaniawala
Ph.D.(Human Genetics)
Consulting Geneticist